

DR. ABHIK ROY

EDUCATION

- Western Michigan University
Ph.D. in Program Evaluation
Kalamazoo, MI
Dissertation. Building an Evaluation Model of Academic Advising's Impact on Progression, Persistence, and Retention Within University Settings
- Michigan Technological University
M.S. in Mathematics
Houghton, MI
Thesis. Quotient Rings of the Eisenstein Integers
- West Virginia Wesleyan College
B.S. in Mathematics
Buckhannon, WV
Report. 4-Cell Embedding on a n -genus Torus

PROFESSIONAL EXPERIENCE

2025
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2024

- Associate Research Scientist
Indiana University
Bloomington, IN
 - Used statistical, machine learning, and qualitative approaches to analyze raw data and generate meaningful evaluation findings.
 - Prepared grant proposals outlining objectives, study designs, budgets, and timelines for external-funding submissions.
 - Presented methods, results, and practical implications to stakeholders through slide decks, briefs, and small-group discussions.
 - Produced color-safe charts, dashboards, and infographics that remained readable for viewers with diverse visual abilities.
 - Monitored budgets, schedules, and task lists to keep projects within scope and on time.
 - Collaborated with external partners in participatory sessions, translating evidence into program guidance and policy draft language.

2023
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2016

- Assistant Professor
West Virginia University
Morgantown, WV
 - Authored ten peer-reviewed articles and spoke at more than twenty national conferences.
 - Contributed to a dozen program evaluations, leading several and supporting analysis on others; most were funded by federal grants in education or health.
 - Taught courses in evaluation, research methods, measurement, and survey design to several hundred students, with an emphasis on applied data science.
 - Supported three master's candidates and one doctoral student from proposal to defense, providing ongoing support with research design, coding, and writing.
 - Served on committees at the graduate (master's and doctoral), program, and college levels.; introduced new courses in data visualization and social network analysis.

CONTACT INFORMATION

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EXPERTISE

Program evaluation integrating traditional and machine learning methods

Quantitative, qualitative, and mixed-methods research

R programming

Statistical modeling and inference

Data visualization

Survey methodology spanning design, deployment, and analysis

Interpretive and computational text analysis

Web application development and reporting using



Social network analysis

Technical writing and markup using *L^AT_EX*

2016
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2014

● Data Analyst
University of Kansas

📍 Lawrence, KS

- Merged surveys, interviews, and registrar data to study how advising affected persistence, retention, and graduation.
- Ran needs assessments and participatory evaluations for advising, student-life, and support units.
- Distributed questionnaires to 300 students and staff; response rates fell between 47% and 91%, depending on the project.
- Built predictive and survival models that flagged at-risk students, supporting tactics that raised retention by as much as five percentage points.
- Synthesized findings from Delphi panels, interviews, survey metrics, and time-to-event analysis into a unified framework for evaluating advising quality.
- Automated data pulls from the Oracle warehouse and refreshed Tableau dashboards with new R scripts.

PUBLICATIONS

- 2024 ● Dey, K., Rahman, M. T., **Roy, A.**, Pyrialakou, V. D., Martinelli, D., Fraustino, J. D., Deskins, J., Rambo-Hernandez, K. E., & Plein, L. C. (2024). Experiences and perceptions of engineering students towards a cross-disciplinary course using sentiment analysis. *Journal of Civil Engineering Education*, 150(3). <https://doi.org/10.1061/jceecd.eieng-1976>
- 2023 ● Kale, U., Kookan, A., Yuan, J., & **Roy, A.** (2023). Teaching science via computational thinking? Enabling future science teachers' access to computational thinking. *Contemporary Issues in Technology and Teacher Education*, 23(3). <https://citejournal.org/volume-23/issue-3-23/science/teaching-science-via-computational-thinking-enabling-future-science-teachers-access-to-computational-thinking>
- Curtis, R., **Roy, A.**, Lewis, N., Dooty, E. N., & Mikalik, T. (2023). Program evaluation standards for utility facilitate stakeholder internalization of evaluative thinking in the West Virginia Clinical Translational Science Institute. *Journal of MultiDisciplinary Evaluation*, 19(43), 49–65. <https://doi.org/10.56645/jmde.v19i43.831>
- 2022 ● Kale, U., Yuan, J., & **Roy, A.** (2022). Thinking processes in Code.org: A relational analysis approach to computational thinking. *Computer Science Education*, 33(4), 545–566. <https://doi.org/10.1080/08993408.2022.2145549>
- 2021 ● **Roy, A.**, & Rambo-Hernandez, K. E. (2021). There's so much to do and not enough time to do it! A case for sentiment analysis to derive meaning from open text using student reflections of engineering activities. *American Journal of Evaluation*, 42(4), 559–576. <https://doi.org/10.1177/1098214020962576>
- 2020 ● Kale, U., **Roy, A.**, & Yuan, J. (2020). To design or to integrate? Instructional design versus technology integration in developing learning interventions. *Educational Technology Research and Development*, 68(2473–2504). <https://doi.org/10.1007/s11423-020-09771-8>
- 2018 ● Rambo-Hernandez, K. E., **Roy, A.**, Morris, M. L., Hensel, R. A. M., Schwartz, J. C., Atadero, R. A., & Paguyo, C. (2018, April). Using interactive theater to promote inclusive behaviors in teams for first-year engineering students: A sustainable approach. Paper presented at the 2018 CoNECD - The Collaborative Network for Engineering and Computing Diversity Conference, Crystal City, Virginia. <https://doi.org/10.18260/1-2-29592>

- Richards-Babb, M., Curtis, R., Ratcliff, B., **Roy, A.**, & Mikalik, T. (2018). General chemistry student attitudes and success with use of online homework: Traditional-responsive versus adaptive-responsive. *Journal of Chemical Education*, 95(5), 691–699. <https://doi.org/10.1021/acs.jchemed.7b00829>
- Miller, T. A., Carver, J. S., & **Roy, A.** (2018). To go virtual or not to go virtual, that is the question: A comparative study of face-to-face versus virtual laboratories in a physical science course. *Journal of College Science Teaching*, 48(2), 59–67. <https://www.jstor.org/stable/26616271>
- 2014 ● Davis, J. D., Smith, D. O., **Roy, A. R.**, & Bilgic, Y. K. (2014). Reasoning-and-proving in algebra: The case of two reform-oriented U.S. textbooks. *International Journal of Educational Research*, 64, 92–106. doi:10.1016/j.ijer.2013.06.012
- 2012 ● **Roy, A. R.**, Hobson, K. A., & Coryn, C. L. S. (2012). What's in a Scriven number? *Journal of MultiDisciplinary Evaluation*, 8(19), 41–45. <https://doi.org/10.56645/jmde.v8i19.372>



BOOK CHAPTERS

- 2017 ● **Roy, A. R.** (2017). Social network analysis: Finding meaning in connections. In T. C. Ahern (Ed.), *Social media*. Hauppauge, NY: Nova Science.

GRANTS

- 2024 ● Mining for Meaning: Uniting Machine Learning and Qualitative Research to Simulate Raters in Text Analysis
National Science Foundation: NSF 19-575: Methodology, Measurement, and Statistics (MMS) - Pending  *Bloomington, IN*
 Roy, A.R. - Principal investigator



INVITED CONTRIBUTIONS

- 2012 ● What is a Scriven number?
The American Evaluation Association Newsletter
 Roy, A.R., Hobson, K.A., & Coryn, C.L.S.



EVALUATIONS



ACTIVE

Current
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2023

Lead Program Evaluator & Methodologist - Healthy Start Initiative ([HRSA-19-049](#) Total Award: \$5,470,000)

West Virginia University Research Corporation

Morgantown, WV

- Created and validated survey instruments to assess participant experiences, collecting data from more than 50 individuals.
- Conducted evaluation studies of program activities and authored the evaluation section of the HRSA annual report, as well as a related federal funding renewal proposal.
- Designed and led program-level evaluations, including process, monitoring, and impact studies.
- Developed and implemented a mixed-methods study to analyze longitudinal and cross-sectional data, focusing on smoking cessation among mothers and the experiences of new or distant caregivers and fathers.

Current
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2022

Program Evaluator & Mixed Methods Analyst - Teaching Science with Computational Thinking: Preparing Preservice Elementary Educators of the Future STEM Workforce ([2019-NSF 2142274](#) Total Award: \$294,958.00)

West Virginia University

Morgantown, WV

- Administered process evaluations for all program activities.
- Created and implemented a longitudinal mixed-method study to analyze more than 50 open-ended survey responses and six interview transcripts, using methods such as HDBSCAN/t-SNE, k-Means, and PCA alongside thematic and content analyses.
- Developed and validated survey instruments to track progress and evaluate the impact of program activities.
- Prepared detailed internal evaluation reports and developed summary documents for federal reporting.



COMPLETED

2022
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2017

Community Program Evaluator & Data Scientist - WVCTSI: West Virginia Clinical and Translational Science Institute ([2017-NIH 2U54GM104942-02](#) Total Award: \$20,000,000)

West Virginia University

Morgantown, WV

- Advised multiple aspiring graduate students in social data science as they developed and completed independent research projects.
- Analyzed datasets using multiple guiding the direction and activities of eight distinct programs.
- Authored quarterly reports as well as internal and external annual evaluation documents, disseminated both in print and through interactive formats developed in Rmarkdown.
- Created data visualizations and developed Shiny applications to support internal and public data exploration, including social network analysis for research collaborations, integration of NCBI API data with WVCTSI grant records, and dissemination of findings and practice changes within and beyond West Virginia.
- Designed and disseminated tailored Qualtrics surveys reaching over 5,000 individuals, with user experience and visual design enhanced through HTML, CSS, and JavaScript.
- Led local and multi-site evaluation studies that informed policies across five core medical research and health outreach efforts.

2020
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2021

Program Evaluator - Appalachian Gerontology Experiences - Advancing Diversity in Aging Research ([2020-NIH 1R25AG059558-01A1](#) Total Award: \$678,000)

West Virginia University

Morgantown, WV

- Developed and distributed two customized interactive surveys using Qualtrics, with user experience and visual design enhanced through HTML, CSS, and JavaScript, to collect responses on programmatic activities from a targeted group of 24 students.
- Led three distinct evaluative studies focusing on student efficacy, engagement, and motivation
- Produced an external evaluation summary for federal reporting.

2020
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2022

Research Methods Advisor & Specialist - Research Initiative: A Holistic Cross-Disciplinary Project Experience as a Platform to Advance the Professional Formation of Engineers ([2019-NSF 1927232](#) Total Award: \$200,000)

West Virginia University

Morgantown, WV

- Conducted longitudinal studies on the experiences of 30 undergraduate students in specialized interdisciplinary courses that integrated social science and engineering, using both surveys and focus group discussions.
- Mentored engineering faculty members and graduate students in the implementation of research methodologies.

2018
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2017

Program Evaluator - Stepping UP with Avenue: Progress Monitoring: A Software Suite Helping Teachers Improve Literacy Progress For Deaf/Hard Of Hearing Students (2017-ED H327S170012 Total Award: \$2,470,440)
Pennsylvania State University

State College, PA

- Evaluated digital tools developed for assessing and engaging individuals who are deaf or hard of hearing.
- Led a multi-site evaluation that examined program delivery, participant engagement, and outcomes.
- Produced a detailed external evaluation brief for federal reporting.

2018
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2017

Lead Program Evaluator - Cultivating Inclusive Identities of Engineers and Computer Scientists: Expanding Efforts to Infuse Inclusive Excellence in Undergraduate Curricula (2017-NSF 1725880 Total Award: \$2,000,000)
West Virginia University

Morgantown, WV

- Applied longitudinal NLP text mining techniques such as concordance, LDA topic modeling, and sentiment analysis to analyze and summarize feedback from over 3,000 first-year engineering students concerning grant-related class activities.
- Carried out longitudinal surveys with over 50 faculty participants to understand expectations, gather input for improvement, and track changes in DEI attitudes and perceptions.
- Created a wide range of static and interactive data visualizations for stakeholder exploration, internal and external reporting, and use in presentations and publications.
- Contributed to multiple journal articles and academic conference presentations.
- Performed personnel evaluations of principal investigators to support broader program assessment efforts.
- Prepared internal evaluation reports and authored the evaluation section of the annual report submitted to the NSF.

2020
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2017

Program Evaluator - GAUSSI: Generating, Analyzing, and Understanding Sensory and Sequencing Information: A Trans-Disciplinary Graduate Training Program in Biosensing and Computational Biology (2017-NSF 1450032 Total Award: \$3,013,779)
Colorado State University

Fort Collins, CO

- Carried out 32 interviews and focus groups, both cross-sectional and longitudinal, using unstructured and semi-structured formats to explore the experiences of students and faculty.
- Conducted process evaluations for multiple grant-funded programs, resulting in improved member tracking, greater program efficiency, and increased participant satisfaction.
- Created and validated instruments to assess students' ability to communicate research findings to a general audience.
- Designed semi-annual adaptive and interactive Qualtrics surveys enhanced with HTML/CSS/JavaScript, securing feedback from over 100 students and faculty with a 95% response rate.
- Generated data visualizations for assessment and longitudinal studies, bolstering inferential statistical analyses to identify trends and support programmatic enhancements, retention strategies, and satisfaction initiatives.
- Produced internal evaluation reports and prepared summaries for use in external communications and federal reporting.



PRESENTATIONS

2022

Let's get sentimental: Machine learning aided data analysis for large qualitative data sets
American Evaluation Association Annual Conference

New Orleans, LA

Seidel, T., Ferguson, C.F., & Roy, A.R.

2022

Best of both worlds: Affordances of mixing machine learning and qualitative content analysis
American Educational Research Association Annual Meeting

San Diego, CA

Roy, A.R., Ferguson, C.F., Curtis, R., & Babb-Richards, M.

2020

These aren't random words just strung together?: Using machine learning and pretty visualizations to discover topics in articles.
American Evaluation Association Annual Conference

virtual

Roy, A.R.

- 2019 ● Little fish in a big pond, only fish in a little pond: How roles shape our identities as evaluators.
American Evaluation Association Annual Conference 📍 Minneapolis, MN
Loomis, D.L., Mikalik, T.L., Curtis, R., **Roy, A.R.**, & Bernstein, M.
- 2019 ● Evolving program logic models to meet shifting program needs: The case of WV Clinical Translational Science Institute.
American Evaluation Association Annual Conference 📍 Minneapolis, MN
Curtis, R., **Roy, A.R.**, Bernstein, M, Loomis, D.L., & Mikalik, T.L.
- 2019 ● The value of external evaluators when building clinical translational research infrastructure.
American Evaluation Association Annual Conference 📍 Minneapolis, MN
Curtis, R., **Roy, A.R.**, Bernstein, M, Loomis, D.L., & Mikalik, T.L.
- 2019 ● Using associated networks to evaluate content within courses.
American Evaluation Association Annual Conference 📍 Minneapolis, MN
Roy, A.R., Kale, U, & Yuan, J.
- 2019 ● Why is it that writers write but fingers don't fmg? Using machine learning and lexemes to make sense of nonsense.
American Evaluation Association Annual Conference 📍 Minneapolis, MN
Roy, A.R., Curtis, R., Mikalik, T.L., Loomis, D.L., & Bernstein, M.
- 2019 ● Iscovering the underlying meaning behind *get me off your f***ing mailing list?* and most other narratives.
American Evaluation Association Annual Conference 📍 Minneapolis, MN
Roy, A.R., Curtis, R., Mikalik, T.L., Loomis, D.L., & Bernstein, M.
- 2019 ● Assessing for improvement: The use of artificial intelligence to uncover potential differential impact of assignments.
American Evaluation Association Annual Conference 📍 Toronto, CN
Roy, A.R. & Rambo-Hernandez, K.
- 2018 ● That's a pretty picture of dots and lines but what does it mean?: A Q&A session with the Social Network Analysis TIG leaders.
American Evaluation Association Annual Conference 📍 Cleveland, OH
Roy, A.R., Durland, M.M., Woodland, R., & Phillips, G.
- 2018 ● Navigating buy-in and shifting evaluation needs over time in NIG Clinical Translational Research Award.
American Evaluation Association Annual Conference 📍 Cleveland, OH
Curtis, R., **Roy, A.R.**, & Mikalik, T.L.
- 2018 ● Using a mixed methods evaluation to discover how an interactive theater based model stimulates inclusive behaviors in engineering.
American Evaluation Association Annual Conference 📍 Cleveland, OH
Roy, A.R., Rambo-Hernandez, K., Hensel, R.A., & Morris, M.L.
- 2018 ● Collaboration evaluation: Using social network analysis to reveal an active undiscovered network.
American Evaluation Association Annual Conference 📍 Cleveland, OH
Roy, A.R., Curtis, R., & Mikalik, T.L.

2018	Examining the past and looking forward: The future of evaluation theory and use. <i>American Evaluation Association Annual Conference</i> Roy, A.R. & Hobson, K.A.	📍 Cleveland, OH
2017	Transforming graduate STEM Education: A theory-driven evaluation of the GAUSSI National Science Foundation Research Training (NRT) Program. <i>American Evaluation Association Annual Conference</i> Roy, A.R., Hernandez, P.A., Chen, T., & Paguyo, C.	📍 Washington, DC
2017	Program evaluation for everyone! - Constructing an online foundational course for capacity building using theorists as a focus. <i>American Evaluation Association Annual Conference</i> Roy, A.R. & Curtis, R.P.	📍 Washington, DC
2017	Three stages down! Exploring the criteria for the next generation of evaluation theorists through social network analysis. <i>Hawaii-Pacific Evaluation Association Annual Conference</i> Roy, A.R. & Hobson, K.A.	📍 Kane'ohe, HI
2017	Content in the background: Using evaluation theorists as the principal motivator for foundational evaluation courses. <i>Hawaii-Pacific Evaluation Association Annual Conference</i> Roy, A.R. & Curtis, R.P.	📍 Kane'ohe, HI
2015	Survey says! Students getting tired of surveys. <i>National Academic Advising Association Annual Conference</i> Roy, A.R. & Goetz, H.L.	📍 Las Vegas, NV
2013	Influences of Hierarchical Linear Modeling in evaluation. <i>Aotearoa New Zealand Evaluation Association Annual Conference</i> Hobson, K.A., Roy, A.R. & Coryn, C.L.S.	📍 Auckland, NZ
2012	Survey sample methods: Evaluators' toolbox refreshment. <i>American Evaluation Association Annual Conference</i> Hobson, K.A., Roy, A.R. & Coryn, C.L.S.	📍 Minneapolis, MN



TEACHING EXPERIENCE



EVALUATION, MEASUREMENT, AND RESEARCH METHODS (2016 - 2023)

2020 2018	<i>Data Visualization</i> West Virginia University 2020, 2018	📍 Morgantown, WV
2017	<i>Educational Psychology</i> West Virginia University	📍 Morgantown, WV

2016	●	<i>Educational Research</i> West Virginia University	📍 Morgantown, WV
2022 2016	●	<i>Introduction to Research</i> West Virginia University 2022, 2018, 2017, 2016	📍 Morgantown, WV
2020 2018	●	<i>Measurement/Evaluation in Educational Psychology</i> West Virginia University 2022, 2020, 2018	📍 Morgantown, WV
2019 2017	●	<i>Mixing Research Methodologies</i> West Virginia University 2022, 2019, 2018, 2017	📍 Morgantown, WV
2023 2017	●	<i>Program Evaluation</i> West Virginia University 2023, 2022, 2021, 2020, 2019, 2018, 2017	📍 Morgantown, WV
2021 2017	●	<i>Social Network Analysis</i> West Virginia University 2021, 2017	📍 Morgantown, WV
2021 2017	●	<i>Statistical Methods 1</i> West Virginia University 2021, 2020, 2019, 2018, 2017	📍 Morgantown, WV
2022 2020	●	<i>Survey Research</i> West Virginia University 2022, 2020	📍 Morgantown, WV

∞ MATHEMATICS (2005 - 2015)

2008	●	<i>Business Calculus</i> Central Michigan University	📍 Mount Pleasant, MI
2009	●	<i>College Algebra</i> Central Michigan University	📍 Mount Pleasant, MI
2012	●	<i>Discrete Mathematics</i> Central Michigan University	📍 Pittsburgh, KS
2014	●	<i>Elementary Statistics</i> Central Michigan University	📍 Pittsburgh, KS

2010 2009	● <i>Foundations of Statistics</i> Central Michigan University 2009, 2010	📍 Pittsburgh, KS
2008 2007	● <i>Intermediate Algebra</i> Central Michigan University 2007, 2008	📍 Mount Pleasant, MI
2007	● <i>Integral Calculus</i> Michigan Technological University	📍 Houghton, MI
2015 2014	● <i>Linear Algebra</i> University of Kansas 2014, 2015	📍 Lawrence, KS
2013	● <i>Mathematical Thinking Grades 6-12</i> Western Michigan University	📍 Kalamazoo, MI
2014 2013	● <i>Mathematics Curriculum Grades 6-12</i> Western Michigan University 2013, 2014	📍 Kalamazoo, MI
2005	● <i>Multivariable Calculus</i> Michigan Technological University	📍 Houghton, MI
2007 2006	● <i>Single Variable Calculus</i> Michigan Technological University 2006, 2007	📍 Houghton, MI

📄 SERVICE

2022 2013	● Associate Editor <i>Journal of MultiDisciplinary Evaluation</i>
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👥 MEMBERSHIPS

2022 2012	● American Evaluation Association
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